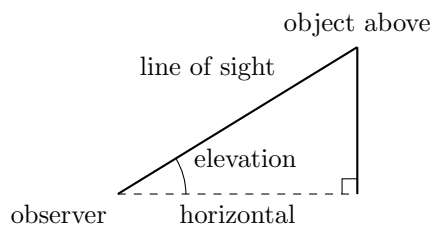


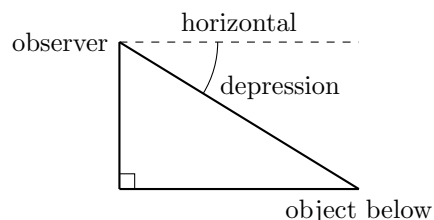
Right Triangles in Elevation and Depression Problems

angles of elevation and depression: finding a side, finding an angle, and two-triangle setups

How the two angles are measured. Both are measured from the *horizontal*, never from the vertical. The reference diagram below shows the labelling every problem on this sheet uses. *No values are shown — use the numbers given in each problem.*



Angle of elevation



Angle of depression

Draw and label the right triangle before you compute. State the ratio you are using, and round each answer as the problem asks. Keep your calculator in degree mode.

1. A surveyor stands on level ground 40 ft from the base of a tree. The angle of elevation from the ground at her feet to the top of the tree is 32° . How tall is the tree, in feet, to the nearest hundredth of a foot?

Answer: _____ ft

2. From the top of a lighthouse 85 m above sea level, the angle of depression to a boat is 12° . How far is the boat from the foot of the lighthouse, measured along the water, to the nearest tenth of a metre?

Answer: _____ m

3. A wheelchair ramp rises 3 ft over a horizontal run of 20 ft. Find the angle of elevation of the ramp, in degrees, to the nearest tenth of a degree.

Answer: _____ degrees

4. A surveyor's eye is 45 m horizontally from the base of a tower, and the top of the tower is 24 m above her eye level. Let θ be the angle of elevation from her eye to the top of the tower. Write $\tan \theta$ as a fraction in lowest terms. (Do not find θ .)

Answer: _____

5. A kite is flying at the end of a taut string 150 ft long. The string makes an angle of elevation of 28° with the ground. How high above the ground is the kite, in feet, to the nearest hundredth of a foot? (Assume the string is held at ground level.)

Answer: _____ ft

6. A plane is cruising at an altitude of 3200 ft. The pilot measures the angle of depression to an airport ahead as 15° . How far is the airport from the point directly beneath the plane, measured along the ground, to the nearest foot?

Answer: _____ ft

7. A student whose eye is 1.6 m above the ground stands 12 m from the foot of a flagpole on level ground. The angle of elevation from her eye to the top of the flagpole is 38° . Find the height of the flagpole above the ground, in metres, to the nearest hundredth of a metre. (The extra measurement is not decoration — decide where it belongs.)

Answer: _____ m

8. A ranger stands at the edge of a cliff 50 m above the sea and looks out at two boats directly in line with the cliff. The angle of depression to the nearer boat is 35° and to the farther boat is 20° . How far apart are the two boats, in metres, to the nearest hundredth of a metre?

Answer: _____ m